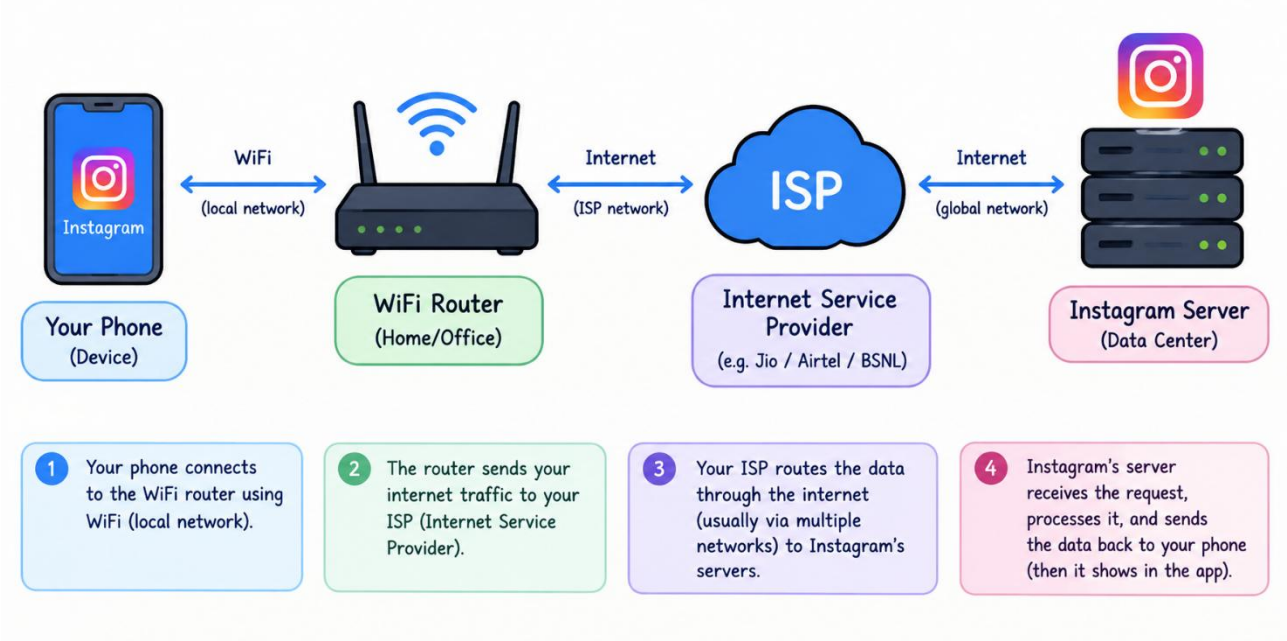
Introduction to Networks

Q1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.

Answer:



Labels and connections:

- **Smartphone:** the device where we open the Instagram app and send requests.
- **WiFi Router:** connects the phone to the local network (LAN) and forwards data to the ISP.
- **ISP (Internet Service Provider):** gives access to the Internet, e.g. Jio or Airtel.
- **Instagram Server:** stores photos, reels and accounts, and sends back the requested data.
- **Phone → Router:** wireless (WiFi).
- **Router → ISP:** cable or fiber.
- **ISP → Server:** Internet backbone (WAN). Data travels in both directions.

Q2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Answer:

No.	Device	Network Type	Reason
1	Smartphone	LAN and WAN	Connects to WiFi router (LAN) and to the Internet via mobile data (WAN)
2	WiFi Router	LAN / WAN	Creates the home LAN and connects it to the ISP (WAN)
3	Laptop	LAN / WAN	Uses WiFi or LAN cable and accesses the Internet
4	Smartwatch	PAN	Connects to the phone through Bluetooth over a very short range
5	Bluetooth earphones	PAN	Pairs with the phone within a few metres

Note: A Smart TV also connects to home WiFi, so it is part of a LAN.

Q3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

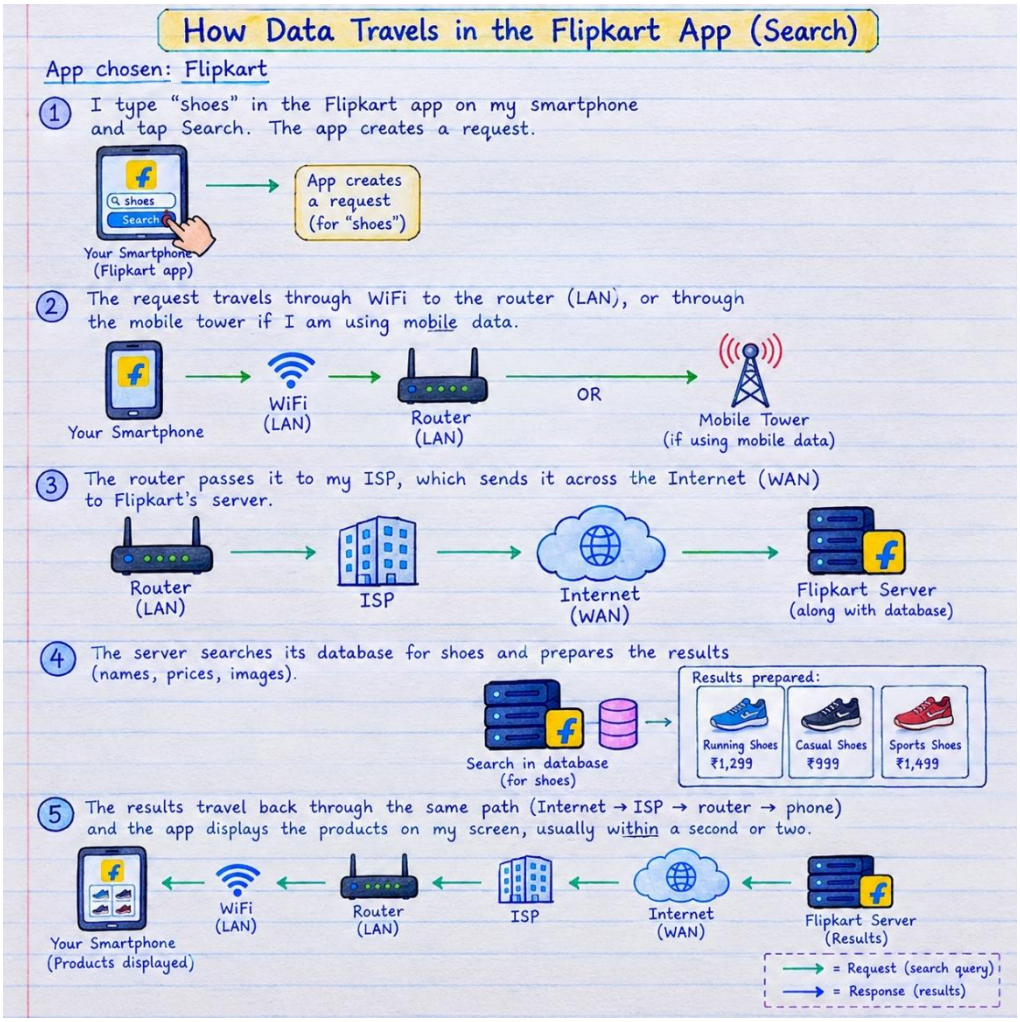
Answer:

- **Internet:** A global public network that anyone can access. Example: public WiFi at a café or railway station, where we can open Google, YouTube or Instagram.
- **Intranet:** A private network used only inside an organization by authorized people. Example: the college portal, accessible only through college WiFi or student login. It is used for attendance, results, notes and the library.

Point	Internet	Intranet
Access	Public, open to everyone	Private, only for members
Size	Worldwide	Limited to one organization
Security	Less secure	More secure (login, firewall)
Example	Google, public WiFi	College portal, office network

Q4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Answer:



App chosen: Flipkart

1. I type "shoes" in the Flipkart app on my smartphone and tap Search. The app creates a request.
2. The request travels through WiFi to the router (LAN), or through the mobile tower if I am using mobile data.
3. The router passes it to my ISP, which sends it across the Internet (WAN) to Flipkart's server.
4. The server searches its database for shoes and prepares the results (names, prices, images).
5. The results travel back through the same path (Internet → ISP → router → phone) and the app displays the products on my screen, usually within a second or two.